

| Assessment | Test | Domain  | Skill                                       | Difficulty                                          |
|------------|------|---------|---------------------------------------------|-----------------------------------------------------|
| SAT        | Math | Algebra | Linear inequalities in one or two variables | Medium <div><div></div><div></div><div></div></div> |

Question

$$y > 14$$
$$4x + y < 18$$

The point  $(x, 53)$  is a solution to the system of inequalities in the  $xy$ -plane. Which of the following could be the value of  $x$ ?

Answer

- A.  $-9$
- B.  $-5$
- C.  $5$
- D.  $9$

Correct Answer: A

Rationale

Choice A is correct. It's given that the point  $(x, 53)$  is a solution to the given system of inequalities in the  $xy$ -plane. This means that the coordinates of the point, when substituted for the variables  $x$  and  $y$ , make both of the inequalities in the system true. Substituting  $53$  for  $y$  in the inequality  $y > 14$  yields  $53 > 14$ , which is true. Substituting  $53$  for  $y$  in the inequality  $4x + y < 18$  yields  $4x + 53 < 18$ . Subtracting  $53$  from both sides of this inequality yields  $4x < -35$ . Dividing both sides of this inequality by  $4$  yields  $x < -8.75$ . Therefore,  $x$  must be a value less than  $-8.75$ . Of the given choices, only  $-9$  is less than  $-8.75$ .

Choice B is incorrect. Substituting  $-5$  for  $x$  and  $53$  for  $y$  in the inequality  $4x + y < 18$  yields  $4(-5) + 53 < 18$ , or  $33 < 18$ , which is not true.

Choice C is incorrect. Substituting  $5$  for  $x$  and  $53$  for  $y$  in the inequality  $4x + y < 18$  yields  $4(5) + 53 < 18$ , or  $73 < 18$ , which is not true.

Choice D is incorrect. Substituting  $9$  for  $x$  and  $53$  for  $y$  in the inequality  $4x + y < 18$  yields  $4(9) + 53 < 18$ , or  $89 < 18$ , which is not true.